BEGINNER

pedro

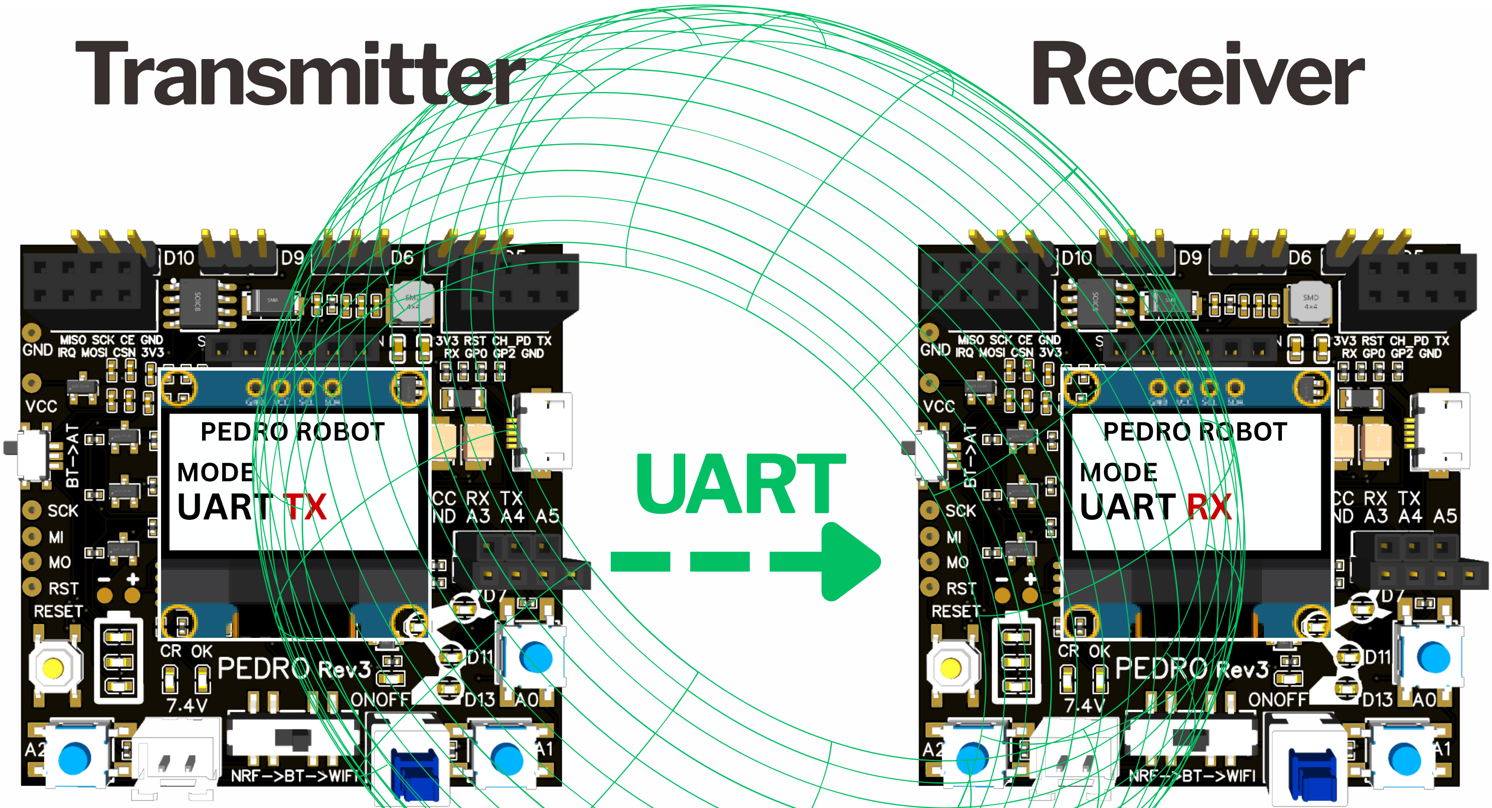
90 min

# STEM Program

## Lesson #8: UART Communication

Transmitter

Receiver



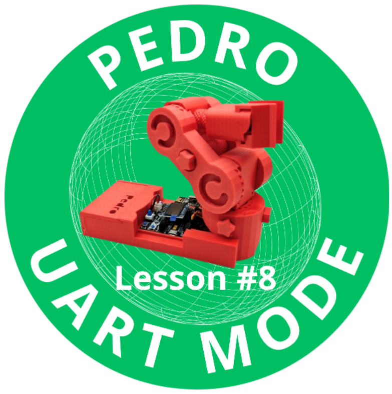

Complete this lesson to earn your Pedro Badge


Add it to your Pedro Passport to track your STEM journey

**YOUR MISSION:** Discover how two robots communicate and exchange information through digital signals using a wired connection.

# Learning Goals

In this lesson, you will discover how two Pedro robots communicate using **UART serial communication**.




By the end of this lesson, you will be able to:

- Explain what **UART communication** is and how it works.
- Identify the **TX**, **RX**, **VCC**, and **GND** pins.
- Correctly wire two Pedro robots for UART communication.

At the end of this lesson, you will unlock your Pedro Badge.
🏅 the Pedro UART Serial Badge

# Teacher Notes

- No prior robotics experience required.
- Duration: 60–90 minutes
- Difficulty: 🟢 Beginner
- Required Materials
    - **2 Assembled** Pedro robot **per student or per group** *(recommended: groups of 2 students)*
    - **OLED Screen 128X64** and **USB cable**
    - **2 jumper wires**
    - **1 computer** with the Arduino IDE installed

**Think:** The **brain icon** encourages students to make careful observations, share their ideas, and engage in discussion by explaining and justifying their answers.

**Learn:** The **light bulb icon** indicates important concepts that students should read carefully. Students are encouraged to ask questions and discuss these topics about the robotics concepts presented in the lesson.

**Make:** The **pencil icon** indicates that students must complete the activity on the page. These activities are designed to reinforce the concepts learned through experimentation, and problem-solving.

# UART Serial Communication

Robots need a way to communicate with computers and electronic components. Before wireless technologies became common, engineers used **wired serial communication** to control and program robotic systems.

Today, **UART (Universal Asynchronous Receiver-Transmitter)** is still widely used in STEM and engineering projects, including:

- robotics
- embedded programming
- automation systems

UART allows Pedro Transmitter and Receiver to exchange information by sending and receiving digital messages.

With UART communication we can:

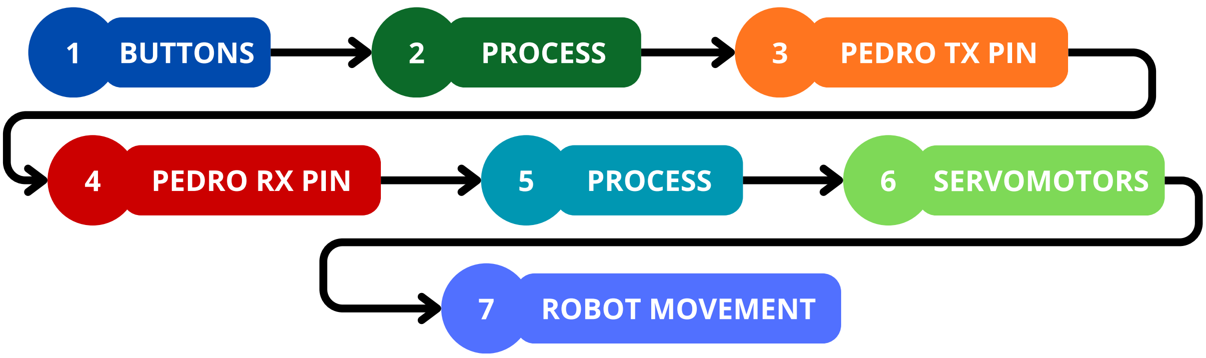
- monitor the robot's behavior
- control movements and actions

UART works like a **conversation between two devices**:

- One device **sends** a digital message.
- The other device **receives and understands** the information.

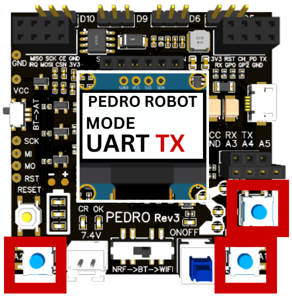
💡 **Note**: UART communication requires a wired connection between the two Pedro robots. Unlike Radio or Bluetooth modes, data is transmitted through the RX and TX pins rather than wirelessly.

**The UART communication chain is:**



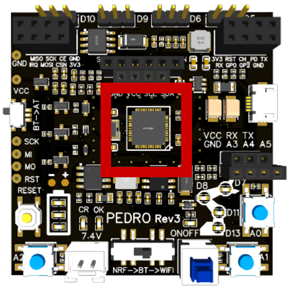
# How UART Mode Works

## 1 BUTTONS



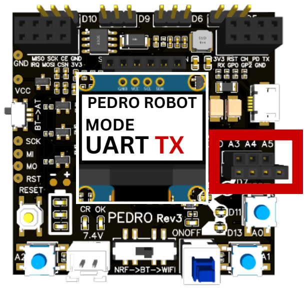

Control the Pedro Transmitter using buttons **A0**, **A1** and **A2**

## 2 PROCESS



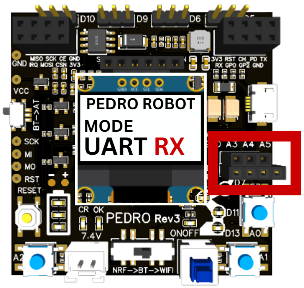

The **microcontroller** processes the button input and sends the command to the TX pin.

## 3 PEDRO TX PIN



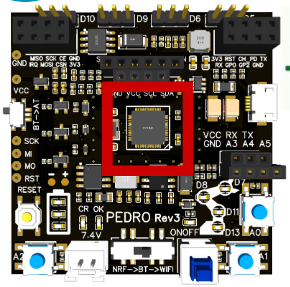

The **TX pin** sends the serial data to the RX pin receiver.

## 4 PEDRO RX PIN




The **RX pin** receives the serial data from the transmitter.

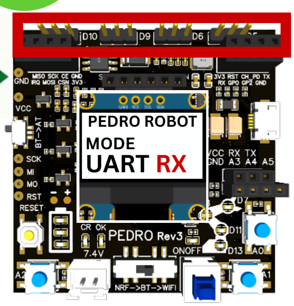
## 5 PROCESS





The **microcontroller** reads the received data and generates the appropriate PWM signal.

## 6 SERVO MOTORS

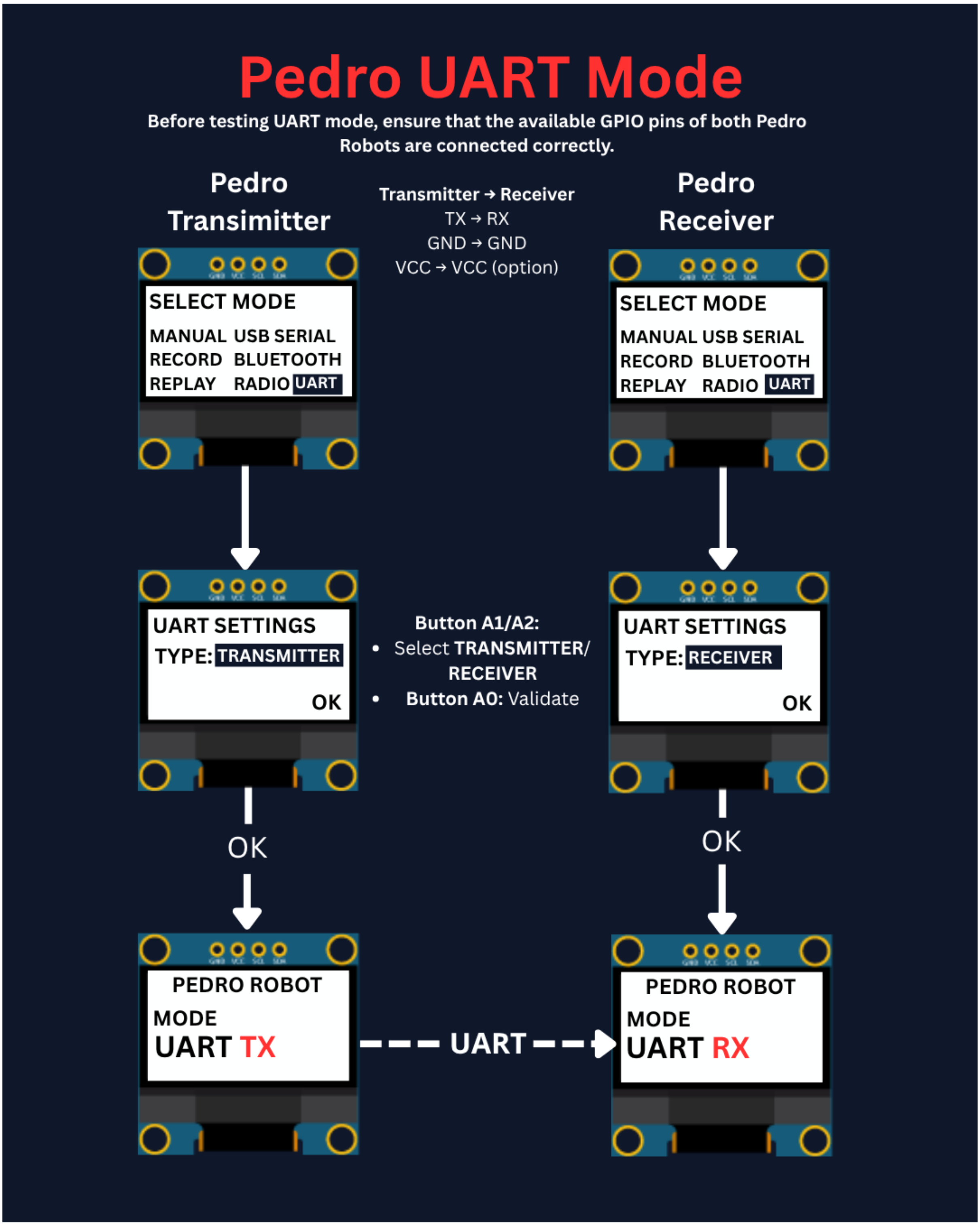



The selected servo receives the **PWM signal** and rotates in the requested direction.

## 7 ROBOT MOVEMENT

The selected servo rotates, causing the robot to perform the requested movement.

## How to Configure UART Mode



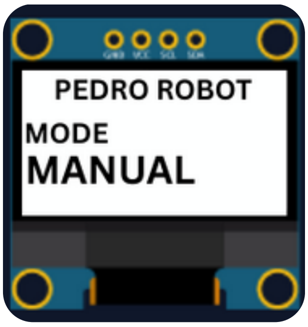
## Board Configuration
## (skip if already configured)

The **Pedro Program** is the core software that powers and controls the robot. It acts as the robot's "brain," managing all behaviors and interactions.

Through this single program, students can access and explore multiple control modes, allowing them to understand how a robot receives instructions, processes them, and translates them into physical actions.

1. **Insert** the Oled Screen into the Pedro board
2. **Connect** the board to your computer with the Micro USB cable.
3. Open **Arduino IDE**
4. **Install the required libraries** from the Library Manager:
   - Tools → Manage Libraries → search **PedroRobot** → Install
   - Tools → Manage Libraries → search **U8glib** → Install
   - Tools → Manage Libraries → search **RF24** → Install
5. Select the port that appear when you connect Pedro robot: **Tools → Port**
6. Select the board: **Tools → Board → Arduino AVR Boards → Arduino Micro**
7. Open the PedroRobot program: **File → Examples → PedroRobot → Pedro**
8. **Compile and upload** the program to your Pedro board.

✅ If everything is correct, Pedro's OLED screen will display **MANUAL MODE**.




Press **A0** and observe what happens. Each time you press **A0**, Pedro selects the next servo.

The LED corresponding to the selected servo lights up, indicating which part of the robot you are controlling. The mapping between the LEDs and the servos is shown below:

- LED D13 → servo D5 (**Base**)
- LED D11 → servo D6 (**Shoulder**)
- LED D8 → servo D9 (**Elbow**)
- LED D7 → servo D10 (**Gripper**)

# Think Before You Build

Before connecting the two Pedro robots, answer the following questions.

**Question 1**

What do you think will happen if the TX pin is connected to another TX pin instead of the RX pin?

✎ Write your prediction:

___

___

___

**Question 2**

What kind of data is transmitted from the TX pin to the RX pin to control the Pedro Receiver?

✎ Explain your answer:

___

___

___

**Question 3**

What do you think will happen if the RX wire is disconnected while Pedro is moving?

✎ Prediction:

___

___

___

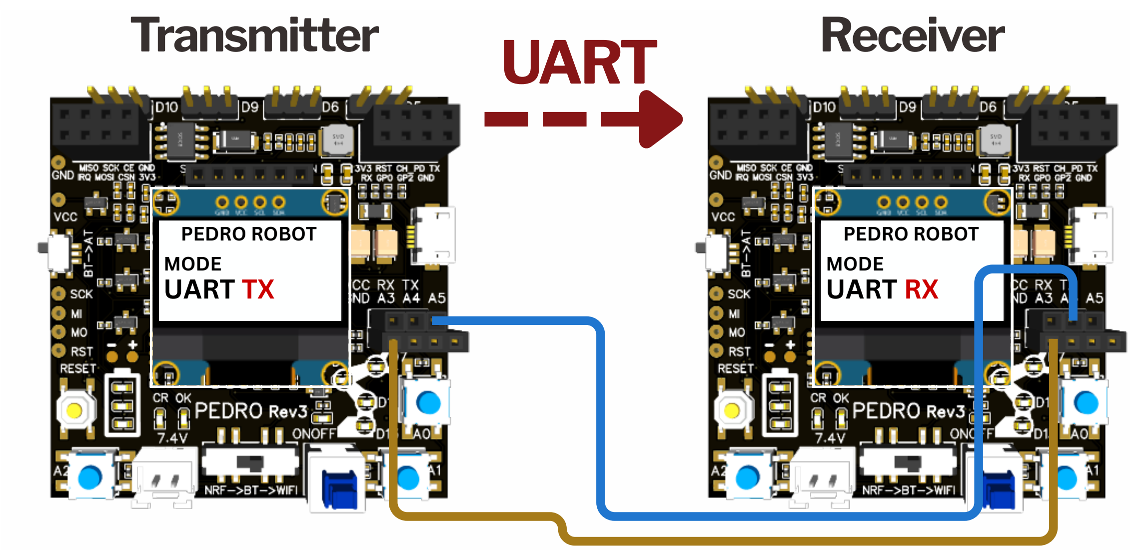
# ✏️ Configuring Pedro in UART mode

**STEP 1:**

UART communication allows two Pedro robots to communicate through the TX and RX pins. Connect the Transmitter and Receiver as shown below.

- **Transmitter → Receiver**
- **TX → RX** (the transmitter's pin must connect to the receiver's RX pin)
- **GND → GND** (a shared ground ensures both interpret voltage levels correctly)

💡 **Note**: Always connect TX → RX (never TX → TX).




**STEP 2:**

Configuring the Pedro Transmitter and Receiver

### UART Transmitter (TX)

1. Power ON Pedro.
2. Enter the **Select Mode** menu.
3. Select **UART Mode.**
4. Press **A0** to confirm.
5. Press **A1** to select the **TRANSMITTER** role.
6. Press **A0** to confirm.
7. Select **OK** then Press **A0** to confirm.

### UART Receiver (RX)

1. Power ON Pedro.
2. Enter the **Select Mode** menu.
3. Select **UART Mode.**
4. Press **A0** to confirm.
5. Press **A1** to select the **RECEIVER** role.
6. Press **A0** to confirm.
7. Select **OK** then Press **A0** to confirm.

## Ready to Communicate

Once both robots are configured, use the **A0**, **A1**, and **A2** buttons on the Pedro **Transmitter** to remotely control the Pedro **Receiver** through the UART (RX/TX) connection.

After connecting the two Pedro robots:

- Press **A1** several times.
- Observe the Receiver.
- Watch how quickly it reacts.

### Discuss

- Is there any visible delay?
- Why do you think UART communication is so fast?
- What is the advantage of using a wired connection instead of radio communication?

## Think Like an Engineer

Suppose you want to build a factory robot.

Which communication would you choose?

- ☐ UART
- ☐ Radio
- ☐ Bluetooth

✎ Explain your choice.

______________________________________________

______________________________________________

## Compare Communication Methods

Complete the table.

| Feature | UART | Radio |
|---|---|---|
| Wired | ☐ | ☐ |
| Wireless | ☐ | ☐ |
| Very Fast | ☐ | ☐ |
| Long Distance | ☐ | ☐ |

Industrial robots often communicate using wired protocols instead of wireless communication.

**Why do you think engineers prefer wired communication in factories?**

Discuss with your classmates.

## Lesson Summary

In this lesson, you learned that:

- **UART** is a wired serial communication protocol.
- Communication requires two wires: **TX** and **RX**.
- The Transmitter sends serial data through its **TX pin.**
- The Receiver receives the data through its **RX pin.**
- The receiving microcontroller processes the command and generates the **PWM signal**.
- The selected servo performs the requested movement.

# Quiz - Check Your Understanding

**1. What is the main purpose of UART Mode on Pedro?**

- [ ] To allow two Pedro robots to communicate through RX and TX pins.
- [ ] To control the robot with Bluetooth.
- [ ] To connect Pedro to a Wi-Fi network.
- [ ] To power another Pedro robot.

**2. Which pin sends serial data from the transmitter?**

- [ ] RX
- [ ] GND
- [ ] VCC
- [ ] TX

**3. Which pin receives serial data on the receiver?**

- [ ] A0
- [ ] PWM
- [ ] RX
- [ ] TX

**4. Why should the GND pins be connected between both Pedro robots?**

- [ ] To replace the TX wire.
- [ ] To power the servos.
- [ ] To provide a common electrical reference.
- [ ] To increase the communication speed.

**5. What happens inside the receiver after the RX pin receives the command?**

- [ ] The robot automatically switches to Bluetooth mode.
- [ ] The RX pin directly moves the servo.
- [ ] The microcontroller process the command and generates a PWM signal.
- [ ] The transmitter receives the message again.

**6. Which buttons control the Pedro Transmitter in UART Mode?**

- [ ] A0, A1 and A2
- [ ] A1 only
- [ ] Power button
- [ ] BT-AT button

**7. Which communication method requires a physical wire between two Pedro robots?**

- ▢ UART
- ▢ Radio
- ▢ Wi-Fi
- ▢ Bluetooth

**8. What is the correct wiring for data transmission?**

- ▢ TX → RX
- ▢ TX → TX
- ▢ RX → RX
- ▢ GND → TX

**9. Which component interprets the PWM signal to produce movement?**

- ▢ Microcontroller
- ▢ USB connector
- ▢ Servomotor
- ▢ RX pin

**10. What is the final result of the UART communication chain?**

- ▢ The robot performs the requested movement.
- ▢ The TX pin stores the command.
- ▢ The receiver powers off automatically.
- ▢ The Bluetooth module changes its name

# 🎯 Congratulations Mission Completed!

You have unlocked: 🏅 the Pedro UART Mode badge
Don't forget to stick your Pedro Badge into your Pedro Passport before starting the next lesson.